report

3. Statistical Methods

3.1 Outcomes and Explanatory Variables

The analysis examined five outcomes across Victorian Local Government Areas (LGAs), including less than moderate physical activity (inactivity), meeting the recommendation of at least 150 minutes of moderate-to-vigorous physical activity (MVPA) per week, obesity, psychological distress, and fair or poor self-rated health. Each outcome was measured as a percentage at the LGA level.

The explanatory variables comprised socioeconomic conditions, population density, and six recreational accessibility indicators. Socioeconomic conditions were measured using the Socio-Economic Indexes for Areas (SEIFA) Index of Relative Socio-economic Advantage and Disadvantage (IRSAD). Population density was included to account for differences in settlement patterns across LGAs. The accessibility indicators included median distance to parks, natural areas, recreational facilities, and indoor facilities, the percentage of facilities that were indoor facilities, and the number of sport types available.

The analysis assessed whether these accessibility indicators provided additional explanatory power beyond socioeconomic conditions and population density.

3.2 Predictor Transformations

Predictor transformations were applied to reduce skewness. Population density and the four distance measures were log-transformed before standardisation. The percentage of indoor facilities was square-root transformed, while SEIFA IRSAD and sport-type count were retained on their original scales before standardisation. All predictors were standardised using z-scores, calculated by subtracting the sample mean and dividing by the sample standard deviation. The indoor-facility share and sport-type count were reverse-coded so that higher values represented a lower indoor-facility share and fewer sport types. SEIFA IRSAD was not reverse-coded, meaning that higher values continued to indicate greater socioeconomic advantage. A park–facility composite was constructed as the arithmetic mean of the standardised park-distance

and facility-distance indicators. This composite was used in Model A to represent these two dimensions jointly.

3.3 Hierarchical Model Specification

Ordinary least squares (OLS) regression models were fitted separately for each of the five outcomes. A sequence of nested and extended models was used to evaluate the explanatory contribution of socioeconomic conditions, population density, and recreational accessibility.

Table 1: Specification of the hierarchical OLS regression models evaluating socioeconomic conditions, population density, and recreational accessibility.

Model	Predictors	Purpose
M0	SEIFA	Establish the socioeconomic baseline
M1	SEIFA + log-transformed population density	Assess the additional contribution of population density
M2	M1 + one accessibility indicator	Evaluate each accessibility indicator individually
A	M1 + park–facility composite + natural-area distance + indoor-facility distance	Evaluate a compact combination of accessibility indicators
B	M1 + park distance + facility distance + natural-area distance + indoor-facility distance	Evaluate the four distance indicators jointly
C	M1 + all six accessibility indicators	Evaluate the full set of accessibility indicators jointly

For each outcome, M0 and M1 were compared to assess whether population density provided additional explanatory power beyond SEIFA IRSAD. Each M2 model added one accessibility indicator to M1, allowing its individual contribution to be assessed. Models A, B and C examined the joint contribution of multiple accessibility indicators using alternative predictor combinations. The extended models were compared with M1 to evaluate whether recreational accessibility improved explanatory power after accounting for socioeconomic conditions and population density.

3.4 Model Comparison and Statistical Inference

Model performance was primarily evaluated using adjusted R^2, which measures the proportion of variation explained by a model while accounting for the number of predictors. For each extended model, the change in adjusted R^2 was calculated relative to M1:

$$R^2_{\text{adj,extended}} - R^2_{\text{adj,M1}}$$

A positive change indicates an increase in adjusted explanatory power, whereas a negative change indicates that the extended model did not improve adjusted (R^2) after accounting for model complexity.

Statistical evidence for incremental contributions was assessed using HC3 heteroskedasticity-robust standard errors and robust F-tests. For M1, the test evaluated the contribution of population density beyond SEIFA IRSAD. For each M2 model, it evaluated the individual accessibility indicator added to M1. For Models A, B and C, joint F-tests assessed whether the accessibility predictors added to M1 were jointly different from zero.

Adjusted (R^2) and HC3 robust tests were interpreted as complementary measures. Adjusted R^2 was used to compare explanatory power, while the robust tests assessed statistical evidence for the additional predictors. Statistical significance was evaluated at the 5% level. These analyses were conducted at the LGA level and were intended to assess associations and explanatory power rather than establish causal effects.

4. Results

4.1 Baseline Model Performance: M0 vs M1

Model M0 used SEIFA as the sole predictor, while Model M1 additionally included population density. The comparison assessed the explanatory power of socioeconomic conditions and whether population density provided additional explanatory power beyond socioeconomic conditions.

Table 2: Baseline model performance across five outcomes. *Note: Δ Adjusted R^2 is calculated as the adjusted R^2 of M1 minus that of M0. HC3 p-values test the additional contribution of population density after controlling for SEIFA.*

Outcome	M0 Adjusted R^2	M1 Adjusted R^2	Δ Adjusted R^2	HC3 p-value
Less than moderate activity	0.617	0.612	-0.005	0.795

Outcome	M0 Adjusted R^2	M1 Adjusted R^2	Δ Adjusted R^2	HC3 p-value
At least 150 min/week MVPA	0.163	0.161	-0.002	0.425
Obesity	0.626	0.645	+0.019	0.121
Psychological distress	0.272	0.317	+0.046	0.029
Fair/poor self-rated health	0.437	0.465	+0.027	0.147

The explanatory power of SEIFA varied substantially across outcomes. Under M0, adjusted R^2 ranged from 0.163 for MVPA to 0.626 for obesity, indicating that socioeconomic conditions explained different proportions of the variation across health and physical activity outcomes.

Adding population density produced mixed results. As shown in Table 2,adjusted R^2 increased for obesity, psychological distress and fair/poor self-rated health, with the largest increase observed for psychological distress (+0.046). However, only psychological distress showed statistically significant evidence of an additional contribution from population density at the 5% level (HC3 p = 0.029). The increases for obesity and fair/poor self-rated health were not statistically significant. For inactivity and MVPA, adjusted R^2 decreased slightly after population density was added.

Overall, population density provided additional explanatory power for psychological distress, but its contribution was not consistently supported across the five outcomes. These results establish M1 as the baseline for evaluating whether recreational accessibility provides further explanatory power.

4.2 Individual Accessibility Contributions: M1 to M2

To examine the individual contribution of recreational accessibility, six M2 models were fitted for each outcome, with each model adding one accessibility indicator to M1. Changes in adjusted R^2 and HC3 robust tests were used to evaluate the incremental explanatory contribution of each indicator.

Table 3: Strongest individual accessibility contributions by outcome. *Note: For each outcome, the indicator shown is the one associated with the highest adjusted R^2 among the six individual M2 models. Δ Adjusted R^2 is calculated relative to M1. The complete results for all 30 M2 models are provided in Appendix [1].*

Outcome	Indicator added	M2 Adjusted R^2	Δ Adjusted R^2	HC3 p-value
Less than moderate activity	Natural-area distance	0.680	+0.068	0.001
At least 150 min/week MVPA	Indoor share (reversed)	0.206	+0.045	0.357
Obesity	Natural-area distance	0.652	+0.008	0.121
Psychological distress	Indoor share (reversed)	0.345	+0.028	0.125
Fair/poor self-rated health	Natural-area distance	0.470	+0.005	0.165

The contribution of individual accessibility indicators differed across outcomes. The clearest improvement was observed for inactivity. Adding natural-area distance increased adjusted R^2 from 0.612 under M1 to 0.680, with statistically significant evidence of an additional contribution (HC3 p = 0.001). The reversed indoor-facility share also improved the adjusted R^2 for inactivity to 0.662 (Δ Adjusted R^2 = 0.050; HC3 p = 0.039). Therefore, both indicators provided evidence of additional explanatory power for inactivity when considered individually. For the other four outcomes, the indicators associated with the highest adjusted R^2 produced smaller improvements, and none had a statistically significant HC3 p-value at the 5% level. For MVPA, the reversed indoor-facility share yielded the largest increase in adjusted R^2 (+0.045), but the evidence was not statistically significant (p = 0.357). Similarly, the largest individual improvements for obesity, psychological distress, and fair/poor self-rated health were not statistically significant.

These findings suggest that individual accessibility indicators provided more substantial additional explanatory power for inactivity than for the other outcomes. However, the results varied across outcomes, and the strongest-performing indicator differed between them. The next section therefore examines whether combining accessibility indicators provides further explanatory power beyond M1.

4.3 Joint Accessibility Contributions: Models A, B and C

To examine whether recreational accessibility indicators provided additional explanatory power when considered jointly, Models A, B and C were compared with the baseline model M1. Each model included SEIFA and population density, while the accessibility indicators were added in different combinations.

Table 4: Joint contribution of accessibility indicators: Models A, B and C versus M1 *Note: Δ Adjusted R^2 is calculated relative to M1. HC3 joint p-values test whether the accessibility predictors added to each model are jointly different from zero. Values reported as <0.001 are based on the unrounded p-values.*

Outcome	Model	Adjusted R^2	Δ Adjusted R^2	HC3 joint p-value
Less than moderate activity	M1	0.612	–	–
	A	0.713	+0.100	<0.001
	B	0.709	+0.097	<0.001
	C	0.712	+0.100	<0.001
At least 150 min/week MVPA	M1	0.161	–	–
	A	0.158	-0.003	0.795
	B	0.155	-0.005	0.666
	C	0.180	+0.020	0.798
Obesity	M1	0.645	–	–
	A	0.644	-0.001	0.479
	B	0.661	+0.016	0.195
	C	0.653	+0.009	0.367
Psychological distress	M1	0.317	–	–
	A	0.336	+0.018	0.347
	B	0.328	+0.010	0.536
	C	0.336	+0.019	0.370
Fair/poor self-rated health	M1	0.465	–	–
	A	0.461	-0.004	0.281
	B	0.454	-0.011	0.411
	C	0.459	-0.005	0.337

The results showed a clear improvement for less than moderate physical activity. All three

combined models increased adjusted R^2 from 0.612 to approximately 0.71. The HC3 joint tests were statistically significant for all three models ($p < 0.001$), indicating that the accessibility indicators provided additional explanatory information beyond socioeconomic conditions and population density. For the other four outcomes, the improvements were smaller and were not statistically significant. Model C produced the highest adjusted R^2 for MVPA (0.180), while Model B performed best for obesity (0.661). For psychological distress, Models A and C achieved similar adjusted R^2 values of 0.336. However, none of these models showed a significant joint contribution from accessibility indicators. For fair or poor self-rated health, all three combined models had lower adjusted R^2 values than M1.

To further compare the three combinations, nested model comparisons were conducted between Models A and B, and between Models B and C.

Table 5: Nested comparisons among combined models. *Note: Δ Adjusted R^2 represents the change in adjusted R^2 when moving from the first model to the second. HC3 p-values assess the additional contribution of the predictors introduced in the larger model.*

Outcome	Comparison	Δ Adjusted R^2	HC3 p-value
Less than moderate activity	A $\rightarrow$ B	-0.003	0.675
	B $\rightarrow$ C	+0.003	0.567
At least 150 min/week MVPA	A $\rightarrow$ B	-0.002	0.384
	B $\rightarrow$ C	+0.025	0.607
Obesity	A $\rightarrow$ B	+0.017	0.071
	B $\rightarrow$ C	-0.008	0.797
Psychological distress	A $\rightarrow$ B	-0.008	0.701
	B $\rightarrow$ C	+0.008	0.216
Fair/poor self-rated health	A $\rightarrow$ B	-0.007	0.864
	B $\rightarrow$ C	+0.006	0.222

The nested comparisons did not show statistically significant improvements when moving from Model A to B or from Model B to C at the 5% significance level. Although adding the separate park and facility distance indicators in Model B increased adjusted R^2 for obesity by 0.017 compared with Model A, this improvement was not statistically significant ($p = 0.071$). Similarly, adding the remaining accessibility indicators in Model C did not significantly improve any outcome.

Overall, the combined results suggest that the explanatory contribution of recreational accessibility varied across health outcomes. The strongest and most consistent evidence was found for inactivity, while the results for the other outcomes provided limited statistical evidence of additional explanatory power.

4.4 Comparison of Accessibility Dimensions

The individual and combined model results provide further insight into which accessibility dimensions were most relevant to each outcome. Among the individual M2 models, natural-area distance and the reversed indoor-facility share showed the largest improvements for inactivity. Natural-area distance also produced the highest adjusted R^2 among individual accessibility models for obesity and fair or poor self-rated health. For MVPA and psychological distress, the reversed indoor-facility share produced the largest individual improvements. However, the combined model results showed that the contribution of accessibility indicators differed across outcomes. Model A performed best for inactivity and psychological distress, Model C performed best for MVPA, and Model B performed best for obesity. No combined accessibility model improved the adjusted R^2 for fair or poor self-rated health.

Table 6: Summary of the strongest individual and combined accessibility models.

Outcome	Strongest individual indicator (M2)	Best combined model	Interpretation
Less than moderate activity	Natural-area distance	Model A	Clear improvement from accessibility indicators, supported by significant individual and joint tests
At least 150 min/week MVPA	Indoor-facility share (reversed)	Model C	Adjusted R^2 increased slightly, but the joint contribution was not significant
Obesity	Natural-area distance	Model B	Some improvement in adjusted R^2, but no significant joint contribution
Psychological distress	Indoor-facility share (reversed)	Model A	Small improvement in adjusted R^2, without significant statistical evidence
Fair/poor self-rated health	Natural-area distance	M1	Accessibility indicators did not improve the adjusted R^2

These findings suggest that no single accessibility dimension consistently provided the greatest

explanatory contribution across all health outcomes. Natural-area distance was particularly relevant in the individual models for inactivity, obesity and self-rated health, while indoor-facility share showed the largest individual improvements for MVPA and psychological distress. Nevertheless, these patterns should be interpreted alongside the joint model tests. Therefore, only inactivity showed clear statistical evidence that accessibility indicators collectively added explanatory power.

Overall, recreational accessibility appeared to be more relevant to differences in physical inactivity across Victorian LGAs than to the other health outcomes examined. The results also suggest that the relationship between recreational accessibility and health may depend on the specific outcome being considered, rather than reflecting a uniform association across all health measures.

Appendix

Appendix A Additional Model Results (M1 vs M2)

Table 7: Incremental explanatory power of accessibility indicators (M1 → M2). *Note: This is the complete results for all 30 M2 models. For each outcome, the indicator shown is the one associated with the highest adjusted R^2 among the six individual M2 models. Δ Adjusted R^2 is calculated relative to M1.*

Outcome	Indicator added	M2 Adjusted R^2	Δ Adjusted R^2	HC3 p-value
Less than moderate activity	Park distance	0.619	+0.007	0.138
Less than moderate activity	Natural-area distance	0.680	+0.068	0.001
Less than moderate activity	Facility distance	0.621	+0.009	0.162
Less than moderate activity	Indoor-facility distance	0.630	+0.017	0.071
Less than moderate activity	Indoor share (reversed)	0.662	+0.050	0.039
Less than moderate activity	Sport diversity (reversed)	0.609	-0.004	0.701
At least 150 min/week MVPA	Park distance	0.150	-0.011	0.908
At least 150 min/week MVPA	Natural-area distance	0.175	+0.015	0.300
At least 150 min/week MVPA	Facility distance	0.156	-0.005	0.440

Outcome	Indicator added	M2 Adjusted R^2	Δ Adjusted R^2	HC3 p-value
At least 150 min/week MVPA	Indoor-facility distance	0.154	-0.007	0.667
At least 150 min/week MVPA	Indoor share (reversed)	0.206	+0.045	0.357
At least 150 min/week MVPA	Sport diversity (reversed)	0.166	+0.005	0.443
Obesity	Park distance	0.641	-0.003	0.630
Obesity	Natural-area distance	0.652	+0.008	0.121
Obesity	Facility distance	0.645	0.000	0.297
Obesity	Indoor-facility distance	0.641	-0.004	0.698
Obesity	Indoor share (reversed)	0.640	-0.004	0.765
Obesity	Sport diversity (reversed)	0.643	-0.001	0.442
Psychological distress	Park distance	0.315	-0.002	0.347
Psychological distress	Natural-area distance	0.308	-0.009	0.927
Psychological distress	Facility distance	0.322	+0.005	0.152
Psychological distress	Indoor-facility distance	0.319	+0.002	0.355
Psychological distress	Indoor share (reversed)	0.345	+0.028	0.125
Psychological distress	Sport diversity (reversed)	0.308	-0.009	0.973
Fair/poor self-rated health	Park distance	0.465	+0.001	0.301
Fair/poor self-rated health	Natural-area distance	0.470	+0.005	0.165
Fair/poor self-rated health	Facility distance	0.462	-0.002	0.439
Fair/poor self-rated health	Indoor-facility distance	0.458	-0.007	0.915
Fair/poor self-rated health	Indoor share (reversed)	0.464	-0.001	0.297
Fair/poor self-rated health	Sport diversity (reversed)	0.469	+0.005	0.229